

Question number	Answer	Additional guidance	Mark
7(a)(i)	An answer that includes four of the following points: • increase in sucrose concentration leads to increase in dopamine release (1) • so leading to increase in desire of rats to eat (1) • {relatively few results to make conclusion / overlapping error bars so results not significant} (1) • the results show a correlation not necessarily causation (1) • relevant comment about methodology of study(1)	accept positive correlation {no significant difference / decreased confidence in conclusion} due to overlapping {error / SD} bars low {reliability / validity} of data due to large error bars e.g. number of rats / gender / age / time / how did they test desire to eat	(4)

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7(a)(ii)	A calculation showing the following steps: • correct calculation of $\sum(x^1-x)^2$ (1) • correct calculation of s to 2 significant figures (1)	<table><tr><th>result (x_1)</th><th>$x_1 - \bar{x}$</th><th>$(x_1 - \bar{x})^2$</th></tr><tr><td>45</td><td>-15</td><td>225</td></tr><tr><td>63</td><td>3</td><td>9</td></tr><tr><td>74</td><td>14</td><td>196</td></tr><tr><td>58</td><td>-2</td><td>4</td></tr><tr><td></td><td>$\sum(x_1 - \bar{x})^2 =$</td><td>434</td></tr><tr><td></td><td>n- 1 =</td><td>3</td></tr><tr><td></td><td>S =</td><td>12</td></tr></table>	result (x_1)	$x_1 - \bar{x}$	$(x_1 - \bar{x})^2$	45	-15	225	63	3	9	74	14	196	58	-2	4		$\sum(x_1 - \bar{x})^2 =$	434		n- 1 =	3		S =	12	(2)
result (x_1)	$x_1 - \bar{x}$	$(x_1 - \bar{x})^2$																									
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7(b)	A description that includes the following points: • released into the blood from the adrenal glands/ binds to receptors on target organs(1) • named physiological effect(1) • second named physiological effect(1)	accept binds to receptors on named target organ e.g. increased heart rate/vasodilation/ increased respiratory rate/ dilation of pupils/relaxation of muscle fibres in wall of gut/ glycogen to glucose in liver /	(3)

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7(c)	Description of diagram or graph • correct description from diagram e.g. pancreas secretes {insulin / glucagon} / control of blood glucose involves pancreas, liver and tissue cells • correct description of graphs linking concentration to either a specific time period or meal e.g. glucose concentration increases after a meal / insulin increases after a meal / glucose concentration decreases during the night Explanation / interpretation (max 3) • normal function of the body is dependent on tight controls of blood glucose levels example of what happens when glucose conc out of sync./ diabetes • (chemo)receptors detect a change in blood {glucose/sugar} levels • neural control of pancreatic hormones release • beta cells secrete insulin • {insulin / glucagon} secreted into the blood which travel to {target organ / liver cells / muscle cells} • {insulin/ glucagon} act as transcription factors • {insulin / glucagon} released which attaches to receptors on specific cell membrane • insulin results in {(increased) uptake of glucose (by cells) which is converted to glycogen / glycogenesis /uptake by GLUT4/ increased respiration} • alpha cells secrete glucagon • (glucagon results in) glycogenolysis / hydrolysis of glycogen/breakdown of glycogen to glucose which is released into the blood / gluconeogenesis / other molecules converted into glucose • blood glucose level returns to {set point / normal} 1 statement with an explanation – this must be included if a response is to access the 6th mark • a correct description of reducing {insulin/glucagon} secretion once blood glucose levels have returned to normal / decreased stimulation of pancreas by nerves	(6)

	L1 2 Ds simple descriptions from graph and /or diagram [most using graphs] L2 level 1 plus 2 statements with basic explanations (or 2 statements with 1 basic and 1 detailed explanation) L3 level 2 plus 2 statements with detailed explanation (eg good use of biological terms) To get 6 there has to be one explanation using negative feedback in control of blood glucose concentration	
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